

Object Detection

Object Detection

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Demonstrate **Ultralytics** using Python: Object Detection effects on images, videos, and real-time detection.

1. Model Introduction

Object detection is a task that involves identifying the location and class of objects in images or video streams.

The output of an object detector is a set of bounding boxes that surround objects in the image, along with class labels and confidence scores for each box. If you need to identify objects of interest in a scene but don't need to know the exact location or precise shape of the objects, object detection is a good choice.

2. Target Prediction: Image

Use yolo26n_ncnn_model to predict images in the ultralytics26 folder.

Enter the code folder:

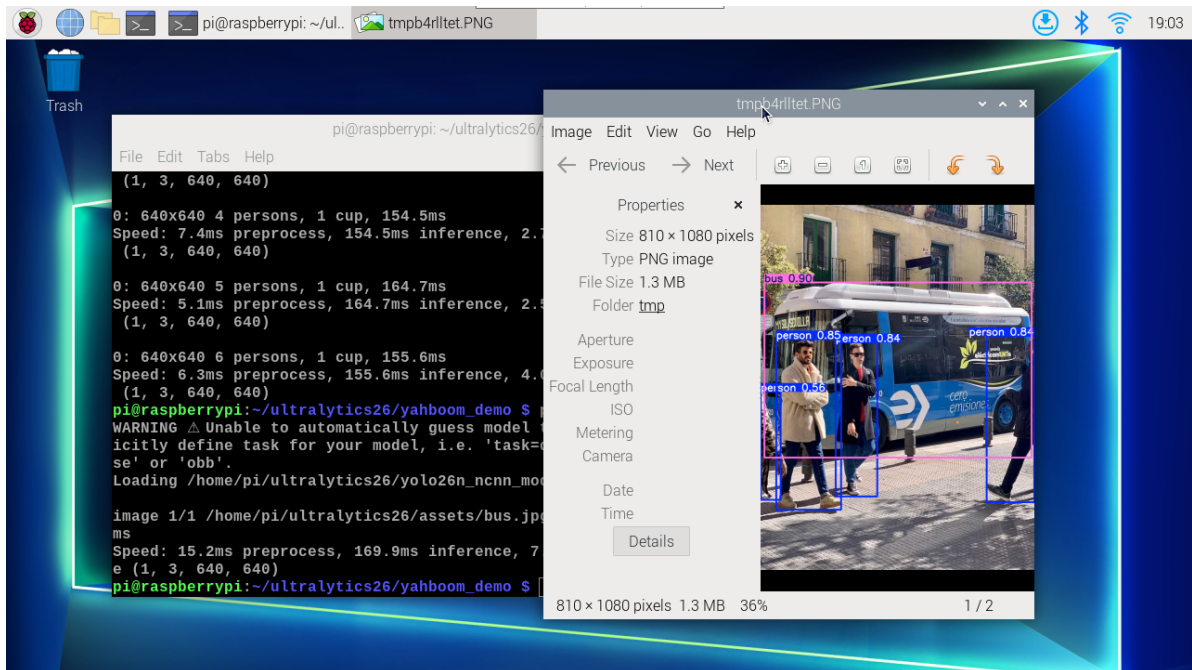
```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 01.detection_image.py
```

Effect Preview

yolo recognition output image location: /home/pi/ultralytics26/output/



Example code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/home/pi/ultralytics26/yolo26n_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n.pt")

# Run batched inference on a list of images
results = model("/home/pi/ultralytics26/assets/bus.jpg") # return a list of Results objects

# Process results list
for result in results:
    boxes = result.boxes # Boxes object for bounding box outputs
    # masks = result.masks # Masks object for segmentation masks outputs
    # keypoints = result.keypoints # Keypoints object for pose outputs
    # probs = result.probs # Probs object for classification outputs
    # obb = result.obb # Oriented boxes object for OBB outputs
    result.show() # display to screen
    result.save(filename="/home/pi/ultralytics26/output/bus_output.jpg") # save to disk
```

3. Target Prediction: Video

Use yolo26n_ncnn_model to predict videos in the ultralytics26 folder (not videos that come with ultralytics).

Enter the code folder:

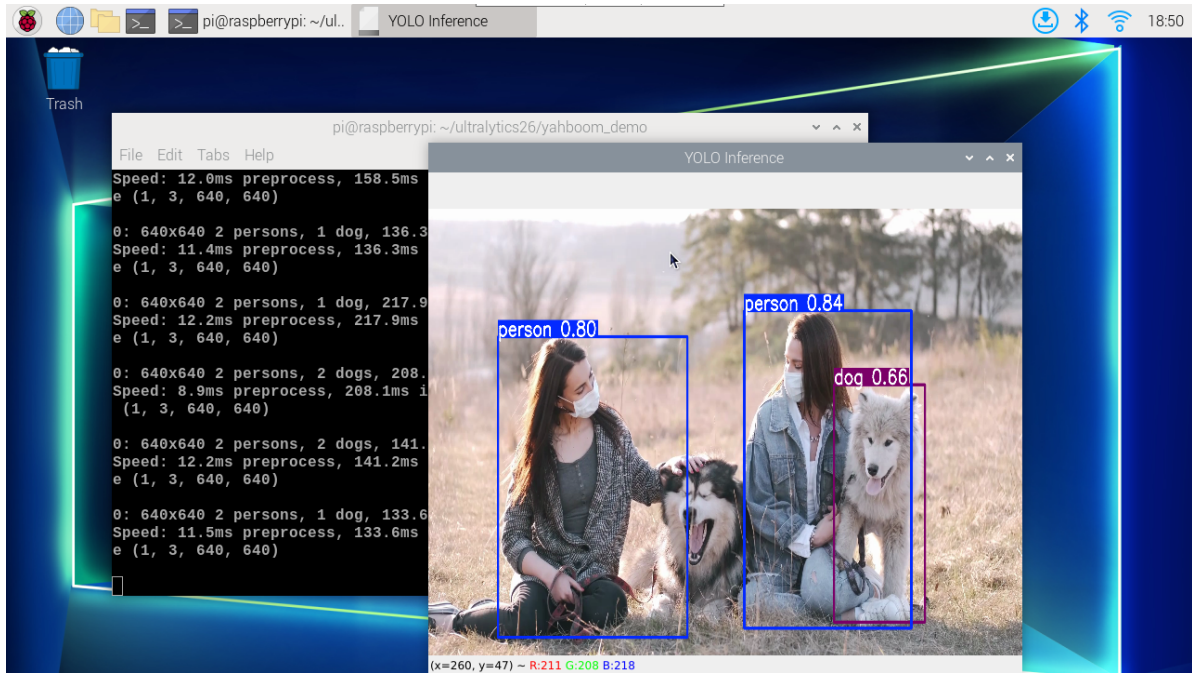
```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 01.detection_video.py
```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/output/



Example code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/pi/ultralytics26/yolo26n_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n.pt")

# Open the video file
video_path = "/home/pi/ultralytics26/videos/people_animals.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/pi/ultralytics26/output/01.people_animals_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v') # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
```

```

annotated_frame = results[0].plot()

# Write the annotated frame to the output video file
out.write(annotated_frame)

# Display the annotated frame
cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

# Break the loop if 'q' is pressed
if cv2.waitKey(1) & 0xFF == ord("q"):
    break
else:
    # Break the loop if the end of the video is reached
    break

# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()

```

4. Target Prediction: Real-time Detection

4.1 USB Camera

Use yolo26n_ncnn_model to predict USB camera footage.

Enter the code folder:

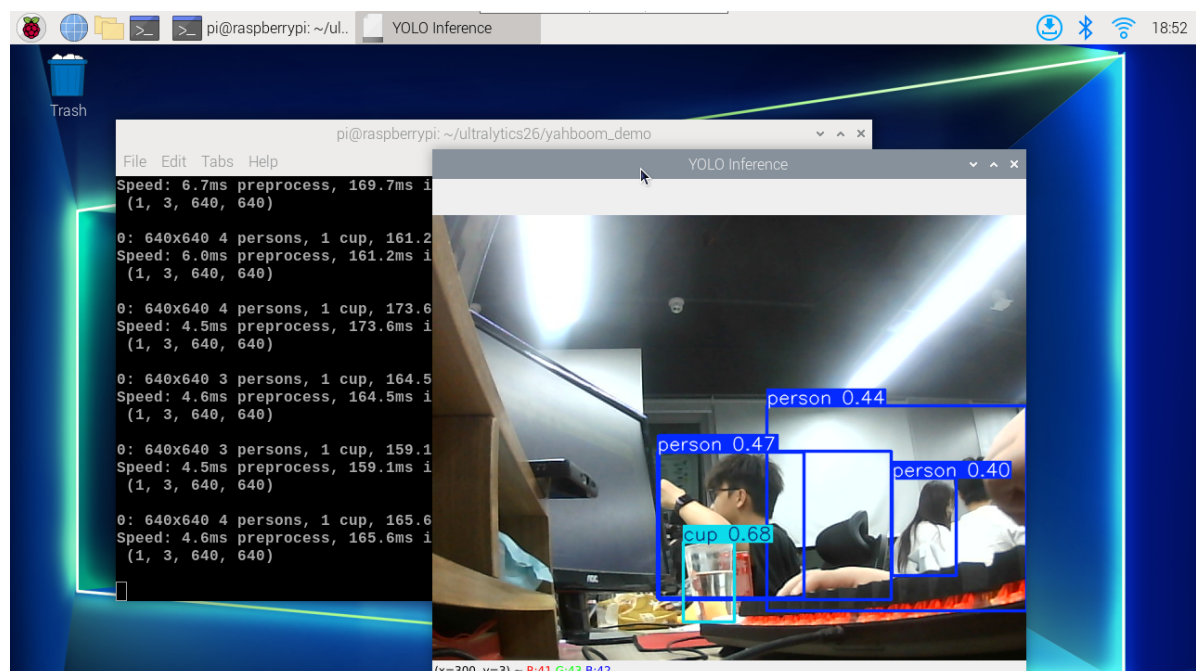
```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 01.detection_camera_usb.py
```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/output/



Example code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/pi/ultralytics26/yolo26n_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n.pt")

# Open the camera
cap = cv2.VideoCapture(0)
cap.set(6, cv2.VideoWriter_fourcc('M', 'J', 'P', 'G'))
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/pi/ultralytics26/output/01.detection_camera_usb.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v') # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break

# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4.2 CSI Camera

Use yolo26n_ncnn_model to predict CSI camera footage.

Enter the code folder:

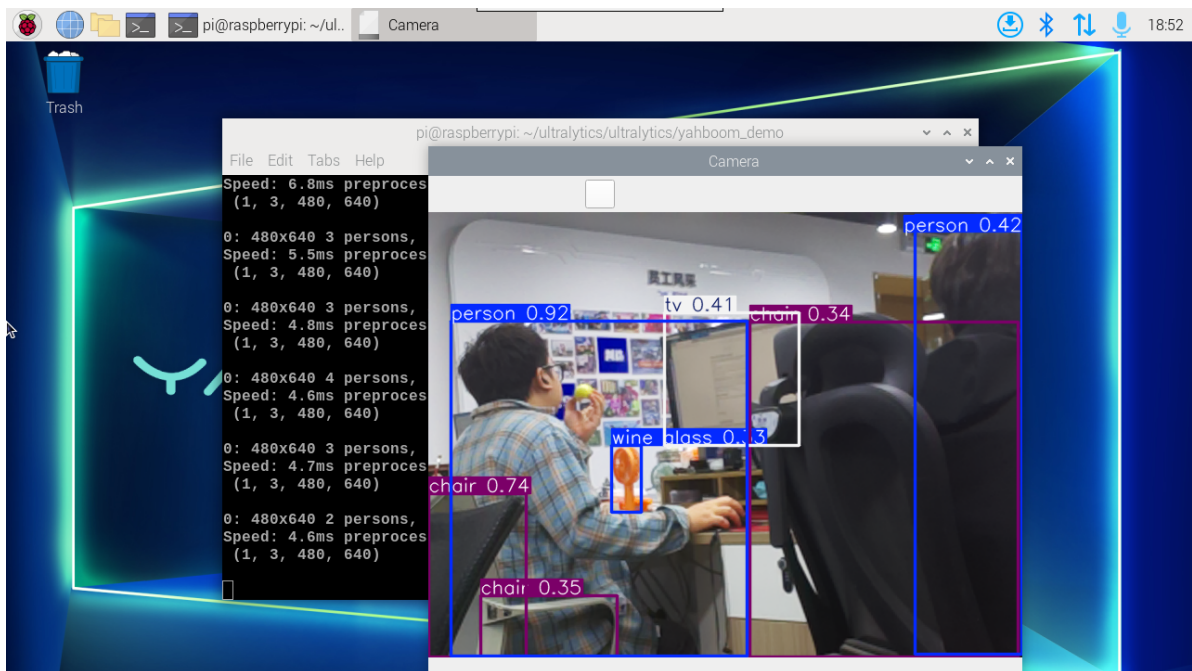
```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 01.detection_camera_csi.py
```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/output/



Example code:

```
import cv2
from picamera2 import Picamera2

from ultralytics import YOLO

# Initialize the Picamera2
picam2 = Picamera2()
picam2.preview_configuration.main.size = (640, 480)
picam2.preview_configuration.main.format = "RGB888"
picam2.preview_configuration.align()
picam2.configure("preview")
picam2.start()

# Load the YOLO26 model
model = YOLO("/home/pi/ultralytics26/yolo26n_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n.pt")

# Set up video output
output_path = "/home/pi/ultralytics26/output/01.detection_camera_csi.mp4"
```



```
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter(output_path, fourcc, 30, (640, 480))

while True:
    # Capture frame-by-frame
    frame = picam2.capture_array()

    # Run YOLO inference on the frame
    results = model(frame)

    # Visualize the results on the frame
    annotated_frame = results[0].plot()

    # Write the frame to the video file
    out.write(annotated_frame)

    # Display the resulting frame
    cv2.imshow("Camera", annotated_frame)

    # Break the loop if 'q' is pressed
    if cv2.waitKey(1) == ord("q"):
        break

# Release resources and close windows
picam2.close()
out.release()
cv2.destroyAllWindows()
```

References

[Object Detection - Ultralytics YOLO Documentation](#)